QCAM — Quality Conformity & Acceptance Module

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OOF® Origin Open Foundation®

Independent Methodological Authority

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Canonical Definition

The Quality Conformity & Acceptance Module (QCAM) governs the assessment,

determination, authorization, recognition, limitation, documentation, and lifecycle continuity of software quality conformity and acceptance.

QCAM establishes the methodological conditions under which software quality

can be recognized as conforming to applicable quality criteria, sufficiently validated for a defined intended use, and accepted by competent authority within explicitly established software, configuration, operational, evidentiary, and temporal boundaries.

It governs the relationships among software quality requirements,

assessable quality criteria, testing and coverage evidence, defect governance states, intended-use conditions, quality conformity determinations, residual quality limitations, acceptance authority, acceptance decisions, acceptance

conditions, and subsequent reassessment.

QCAM distinguishes quality conformity from intended-use fitness, technical

assessment from acceptance authority, acceptance from release authorization, conditional acceptance from unconditional acceptance, and accepted software quality from proof of defect-free software.

It applies to software assessed or accepted by humans, organizations,

automated quality systems, AI coding assistants, autonomous development agents, and software governance infrastructure.

QCAM is technology-independent and does not prescribe a particular quality management framework, certification scheme, testing platform, programming language, or software development methodology.

Its principal output is:

Governed Software Quality Conformity & Acceptance State

Its foundational distinction is:

Quality Acceptance ≠ Proof of Defect-Free Software.

Operational Role

The operational role of QCAM is:

To ensure that software quality conformity and acceptance decisions are based on identifiable quality requirements, applicable criteria, competent testing and defect evidence, demonstrated correspondence with intended use, explicit residual limitations, and legitimate acceptance authority, while preserving the boundaries and continuing validity of each determination.

QCAM establishes the conformity and acceptance governance mechanism of SQS.

It determines:

• What software quality conditions are being assessed

• Which criteria apply

• What evidence supports conformity

• Whether the evidence is sufficient

• Which intended-use conditions have been validated

• Which defects and limitations remain

• Who has legitimate authority to accept the quality state

• What the acceptance decision actually covers

• Which conditions limit or invalidate that decision

QCAM does not independently establish software requirements, execute tests, resolve defects, authorize software release, approve deployment, or determine legal compliance.

Operational Question

Under which defined quality criteria, demonstrated software behavior,

intended-use conditions, evidence requirements, residual limitations, and legitimate authority can a specific software artifact and configuration be recognized as quality-conforming and accepted without treating that acceptance as proof of universal correctness, unrestricted fitness, or authorization for release and deployment?

Position in Architecture Governance Flow

The SLGA™ governance flow is:

Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

Cross-cutting:

Software Lifecycle Control

QCAM operates within:

Software Quality Standard (SQS)

The SQS internal governance flow is:

Quality Requirements → Quality Criteria Definition → Testing & Coverage

Management → Defect Management → Quality Conformity & Acceptance

QCAM governs:

Quality Conformity & Acceptance

It is the fifth Core Module of SQS.

Its principal upstream references are:

Governed Software Quality Requirements State

from:

QRM — Quality Requirements Module

Governed Quality Criteria Definition State

from:

QCDM — Quality Criteria Definition Module

Governed Software Testing & Coverage State

from:

TCMM — Testing & Coverage Management Module

Governed Software Defect State

from:

DFM — Defect Management Module

Relevant supporting references include:

Governed Software Requirements Baseline

from Software Requirements Standard (SRS).

Governed Software Creation State

from Software Creation Standard (SCS).

Governed Software Dependency State

from Software Dependency Standard (SDS).

Governed Software Configuration State

from Software Configuration Standard (SCFS).

Its principal output is:

Governed Software Quality Conformity & Acceptance State

This output supports the parent SQS determination:

Governed Software Quality State

It also provides quality-specific evidence and acceptance references to

downstream lifecycle governance, including Software Release Standard (SRLS) and Software Deployment Standard (SDPS).

QCAM may be invoked repeatedly as software artifacts, quality criteria, configurations, operating conditions, defect states, evidence, or acceptance authority change.

• Governed Space

Software Quality Conformity & Acceptance Governance

This space governs the conditions under which software quality conformity

and acceptance determinations become identifiable, evidence-supported, legitimately authorized, and appropriately bounded.

Its principal dimensions include:

• Quality conformity governance object

- Acceptance governance object
- Software quality assessment identity
- Applicable quality requirements
- Quality criteria applicability
- Quality criteria validity
- Software artifact identity
- Artifact version correspondence
- Configuration correspondence
- Intended-use definition
- Intended-use boundaries
- Operational context
- Quality evidence sufficiency
- Testing evidence correspondence
- Coverage adequacy
- Defect state correspondence
- Known defect assessment
- Residual quality limitations
- Quality conformity determination
- Partial conformity
- Nonconformity
- Conformity uncertainty
- Intended-use validation
- Operational fitness assessment
- Acceptance eligibility
- Acceptance authority
- Acceptance authority scope
- Acceptance decision identity
- Acceptance decision basis
- Acceptance conditions
- Conditional acceptance
- Acceptance exclusions
- Acceptance restrictions
- Acceptance validity
- Acceptance evidence
- Acceptance evidence integrity
- Acceptance conflicts

• Acceptance evidence gaps

• Critical acceptance gaps

• Acceptance continuity

• Acceptance reassessment

• Acceptance suspension

• Acceptance withdrawal

• Acceptance supersession

• Quality governance handoff

• Conformity and acceptance governance states

This governed space remains distinct from testing under TCMM, defect

management under DFM, security conformity under SSS, release authorization under SRLS, and deployment authorization under SDPS.

Quality Conformity Governance Object

Every material quality conformity assessment shall establish a Quality

Conformity Governance Object.

The object represents an identifiable assessment of whether a defined

software artifact and configuration satisfy applicable quality criteria within an established scope.

It should identify:

Assessment Identity: The identifiable conformity assessment.

Software Object: The software being evaluated.

Artifact Identity: The relevant software artifact.

Configuration Reference: The configuration under assessment.

Quality Requirements: The applicable governed requirements.

Quality Criteria: The criteria used for conformity assessment.

Intended Use: The operational purpose relevant to the assessment.

Testing Evidence: The supporting testing and coverage state.

Defect Evidence: The relevant defect governance state.

Residual Limitations: Known unresolved quality conditions.

Conformity Determination: The supported assessment outcome.

Evidence Basis: The competent evidence supporting that outcome.

Governance State: The recognized conformity condition.

The conformity governance object shall remain distinguishable from a

quality acceptance decision.

Quality Conformity Determination ≠ Quality Acceptance Decision.

Quality Acceptance Governance Object

Every material quality acceptance decision shall establish a Quality

Acceptance Governance Object.

The object represents the identifiable decision to accept, conditionally

accept, reject, defer, or otherwise determine the acceptance eligibility of a defined software quality state.

It should identify:

Acceptance Identity: The identifiable decision.

Acceptance Subject: The software artifact and configuration.

Conformity Reference: The relevant quality conformity determination.

Intended-Use Scope: The operational conditions covered.

Acceptance Authority: The competent decision authority.

Decision Basis: The supporting evidence and criteria.

Residual Limitations: Known unresolved quality conditions.

Acceptance Conditions: Any restrictions or prerequisites.

Decision Outcome: The recognized acceptance determination.

Decision Validity: The conditions under which the decision remains

applicable.

Continuity Reference: The relationship to later changes and reassessments.

An acceptance record shall not independently establish that its underlying conformity evidence is competent.

Documented Acceptance ≠ Evidence-Supported Acceptance.

Quality Assessment Purpose

QCAM shall identify the purpose of each material conformity or acceptance

assessment.

Purposes may include:

• Verification of quality requirement satisfaction

• Assessment of software readiness for intended use

• Evaluation of residual quality limitations

• Determination of acceptance eligibility

• Reassessment following software changes

• Review of previously accepted software quality

The assessment purpose shall remain connected to the applicable software

object and quality criteria.

Assessment Purpose ≠ Assessment Outcome.

Quality Assessment Scope

QCAM shall establish the boundaries of each material quality assessment.

Scope may concern:

• Software functions

• Components

• Applications

• Integrated systems

• Artifact versions

• Configurations

• Operating environments

• Intended-use scenarios

• User populations

• Relevant quality characteristics

A conformity determination shall not be generalized beyond its established

scope without competent justification.

Quality Conformity Within Scope ≠ Universal Software Conformity.

Software Object Correspondence

QCAM shall identify the software object to which conformity and acceptance

determinations apply.

The object may be:

• A software component

• An application

• A service

• An embedded software system

• An integrated software product

• A defined software configuration

Acceptance of one software component shall not automatically establish

acceptance of the entire integrated system.

Component Quality Acceptance ≠ System Quality Acceptance.

Artifact Identity Governance

QCAM shall preserve the identity of the software artifact actually assessed.

Relevant information may include:

• Artifact identifier

• Version

• Build reference

• Content integrity reference

• Relevant provenance

• Configuration reference

A declared version label shall not independently establish the identity of

the artifact assessed.

Declared Accepted Version ≠ Verified Accepted Artifact.

Artifact Version Correspondence

QCAM shall preserve the relationship between quality evidence and the

artifact version for which acceptance is sought.

Evidence from one version shall not automatically establish conformity of

another.

Material version differences shall remain explicit.

Quality Acceptance of Version A ≠ Quality Acceptance of Version B.

Configuration Correspondence

QCAM shall identify the software configuration to which conformity and

acceptance apply.

Relevant configuration conditions may include:

• Runtime settings

• Dependency versions

• Feature flags

• Build parameters

• Integration configuration

• Relevant external services

Acceptance of one configuration shall not automatically extend to

materially different configurations.

Accepted Configuration ≠ Every Supported Configuration.

Quality Requirement Correspondence

QCAM shall preserve the relationship between conformity determinations and

governed quality requirements.

Requirements should remain identifiable as:

• Satisfied by competent evidence

• Partially satisfied

• Not satisfied

• Not assessed

• Not applicable

• Unresolved

A declared requirement status shall not independently establish conformity.

Requirement Marked Satisfied ≠ Demonstrated Requirement Satisfaction.

Quality Criteria Applicability

QCAM shall determine which governed quality criteria apply to the assessed

software and intended use.

Applicability may depend on:

• Software purpose

• Quality characteristics

• Operational environment

• User requirements

• Integration conditions

• Relevant limitations

QCAM shall not silently modify criteria to produce a favorable conformity

result.

Quality Criterion Applicability ≠ Quality Criterion Satisfaction.

Quality Criteria Validity

QCAM shall assess whether the criteria used for conformity determination

remain applicable and sufficiently established.

Relevant considerations may include:

• Criterion identity

• Approved interpretation

• Applicable threshold

• Assessment method

• Evidence requirements

• Revision status

• Material limitations

Criteria definition remains governed by QCDM.

QCAM evaluates whether the established criteria are suitable for the conformity determination being made.

Established Criterion ≠ Currently Applicable Criterion.

Intended-Use Definition

QCAM shall identify the intended use against which software quality fitness

is evaluated.

Intended use may concern:

• Operational purpose

• Relevant users

• Required functionality

• Expected workload

• Operating environment

• Critical operating conditions

• Relevant dependencies

A general statement that software is ready shall not substitute for identifiable intended-use conditions.

Declared Intended Use ≠ Demonstrated Intended-Use Fitness.

Intended-Use Boundaries

QCAM shall establish the boundaries of the intended use covered by a

quality assessment.

Boundaries may include:

• Supported functions

• User groups

• Workload ranges

• Operating conditions

• Environmental restrictions

• Data characteristics

• Relevant exclusions

Acceptance shall remain bounded by these conditions.

Accepted Intended Use ≠ Unrestricted Software Use.

Operational Context Governance

QCAM shall preserve the operational context relevant to quality conformity

and acceptance.

Context may include:

• Expected operating conditions

• User interactions

• Integration dependencies

• Data volume

• Resource availability

• Failure conditions

• Recovery expectations

A technically successful assessment in a limited environment shall not

automatically establish operational fitness in materially different conditions.

Technical Test Success ≠ Demonstrated Operational Fitness.

Quality Evidence Sufficiency

QCAM shall determine whether available evidence is sufficient to support the intended conformity claim.

Evidence sufficiency may depend on:

• Applicable criteria

• Testing scope

• Test validity

• Coverage limitations

• Defect states

• Artifact correspondence

• Configuration correspondence

• Intended-use conditions

Evidence may be authentic and technically valid while insufficient for the claimed acceptance scope.

Valid Quality Evidence ≠ Sufficient Quality Evidence.

Testing Evidence Correspondence

QCAM shall preserve correspondence between conformity determinations and

the testing evidence governed by TCMM.

Relevant information may include:

• Tested artifact

• Tested configuration

• Applicable criteria

• Execution evidence

• Verified results

• Coverage scope

• Testing exclusions

• Unresolved testing conditions

A passing test report shall not independently establish quality conformity.

Passing Test Report ≠ Quality Conformity Determination.

Coverage Adequacy Assessment

QCAM shall evaluate whether the testing coverage available is adequate for the quality determination sought.

Adequacy shall be assessed relative to:

• Applicable requirements

• Quality criteria

• Intended use

• Operational consequences

• Relevant failure conditions

• Known coverage gaps

High numerical coverage shall not automatically establish adequacy.

High Test Coverage ≠ Sufficient Quality Evidence.

Defect State Correspondence

QCAM shall preserve correspondence with defect governance states

established by DFM.

Relevant conditions may include:

• Confirmed defects

• Unconfirmed defects

• Corrected defects

• Verified resolutions

• Residual defects

• Deferred defects

• Critical defect gaps

• Unresolved verification evidence

Administrative defect closure shall not independently establish defect

resolution.

Closed Defect Record ≠ Verified Quality Conformity.

Known Defect Assessment

QCAM shall assess the significance of known defects for the relevant

conformity and acceptance determination.

Assessment may consider:

• Defect severity

• Operational impact

• Exposure

• Affected functionality

• Available workarounds

• Residual limitations

• Relevant intended-use conditions

Known defects shall not automatically invalidate every quality acceptance

determination.

Their significance must be assessed against the applicable criteria and acceptance scope.

Known Defect Existence ≠ Automatic Quality Rejection.

Residual Quality Limitations

QCAM shall identify material quality limitations remaining after assessment.

Limitations may include:

• Known defects

• Incomplete coverage

• Unverified operating conditions

• Unsupported configurations

• Performance restrictions

• Reliability uncertainty

• Incomplete intended-use validation

Residual limitations shall remain explicit in the acceptance decision.

Residual Limitation Disclosure ≠ Residual Limitation Acceptance.

Quality Conformity Determination

QCAM shall establish a supported determination of whether applicable

quality criteria have been satisfied.

Possible determinations include:

Conforming: Competent evidence supports satisfaction of the applicable

criteria within the assessed scope.

Partially Conforming: Evidence supports satisfaction of some but not all

applicable criteria.

Nonconforming: Competent evidence demonstrates failure to satisfy one or

more applicable criteria.

Inconclusive: Available evidence does not support a sufficient determination.

Not Assessed: The relevant criteria have not been evaluated.

Evidence-Limited: Material evidence deficiencies prevent competent

interpretation.

Conformity determination shall remain distinct from acceptance authority.

Demonstrated Quality Conformity ≠ Authorized Quality Acceptance.

Partial Quality Conformity

QCAM shall recognize conditions where software satisfies only part of the applicable quality criteria.

Partial conformity may arise from:

• Incomplete testing

• Unresolved quality requirements

• Known defects

• Limited environment coverage

• Conditional performance

• Unverified functionality

Partial conformity shall not be represented as complete conformity.

Partial Quality Conformity ≠ Full Quality Conformity.

Quality Nonconformity

QCAM shall recognize demonstrated failure to satisfy applicable quality

criteria.

Nonconformity shall preserve:

• Criterion identity

• Affected software object

• Supporting evidence

• Relevant defect references

• Operational significance

• Applicable scope

A nonconformity determination shall remain distinguishable from an

administrative decision to reject software acceptance.

Quality Nonconformity ≠ Automatically Rejected Acceptance Decision.

Conformity Uncertainty

QCAM shall preserve uncertainty affecting quality conformity determinations.

Uncertainty may arise where:

• Evidence is incomplete

• Test results conflict

• Coverage is insufficient

• Artifact correspondence is unresolved

• Defect verification remains pending

• Intended-use conditions are insufficiently represented

Uncertainty shall not be silently converted into conformity.

Unresolved Quality Evidence ≠ Demonstrated Conformity.

Intended-Use Validation

QCAM shall assess whether the software quality evidence supports fitness

for the defined intended use.

Validation shall consider:

• Operational purpose

• Applicable quality requirements

• Relevant quality criteria

• Tested scenarios

• Observed behavior

• Defect conditions

• Environmental correspondence

• Residual limitations

Conformity with individual technical criteria shall not automatically

establish fitness for the complete intended use.

Technical Quality Conformity ≠ Intended-Use Validation.

Operational Fitness Assessment

QCAM shall evaluate whether the assessed software quality is sufficiently aligned with the relevant operational conditions.

Operational fitness may depend on:

• Expected workload

• User interaction

• Reliability expectations

• Performance constraints

• Integration behavior

• Recovery requirements

• Environmental conditions

Operational fitness shall remain bounded by available evidence.

Fitness for One Operational Context ≠ Fitness for Every Operational

Context.

Quality Acceptance Eligibility

QCAM shall determine whether a software quality state is eligible for acceptance consideration.

Eligibility may require:

• Identifiable software artifact

• Applicable quality criteria

• Competent conformity assessment

• Relevant intended-use validation

• Known defect assessment

• Residual limitation disclosure

• Sufficient evidence

• Available competent acceptance authority

Eligibility shall remain distinct from an actual acceptance decision.

Acceptance Eligibility ≠ Acceptance Authorization.

Acceptance Authority Governance

QCAM shall preserve the legitimate authority under which quality acceptance

decisions are made.

Authority may derive from:

• Organizational governance

• Assigned quality responsibility

• Contractual arrangements

• Delegated decision authority

• Applicable institutional procedures

Authority shall remain identifiable and appropriately scoped.

Technical expertise alone shall not establish legitimate acceptance

authority.

Quality Assessment Competence ≠ Quality Acceptance Authority.

Acceptance Authority Scope

QCAM shall establish the boundaries of acceptance authority.

Authority scope may concern:

• Software product

• Component

• Artifact version

• Quality characteristics

• Intended use

• Operational context

• Permitted residual limitations

• Decision duration

Authority to accept one quality condition shall not automatically extend to

every software quality condition.

Authority to Accept One Scope ≠ Authority to Accept Every Scope.

Human Acceptance Authority

QCAM shall preserve the role of competent human or organizational authority

where required.

Human acceptance may be necessary where:

• Material operational consequences exist

• Residual limitations require judgment

• Institutional accountability applies

• Delegated authority requires human confirmation

• Automated conclusions remain insufficient

Human involvement shall not automatically establish the competence of the

decision.

Human Approval ≠ Competent Quality Acceptance.

Automated Acceptance Authority

QCAM shall recognize acceptance activities supported or performed by

automated systems.

Automated systems may:

• Aggregate evidence

• Evaluate predefined criteria

• Identify conformity gaps

• Generate acceptance recommendations

• Execute bounded acceptance decisions where legitimately authorized

Automated execution shall remain within explicit authority boundaries.

An AI system's ability to produce an acceptance recommendation shall not

establish authority to approve software quality.

Automated Acceptance Recommendation ≠ Authorized Acceptance Decision.

Acceptance Decision Identity

QCAM shall establish identifiable references for material acceptance

decisions.

A decision should preserve:

• Decision identity

• Decision subject

• Acceptance authority

• Applicable criteria

• Evidence basis

• Intended-use scope

• Decision outcome

• Relevant conditions

• Decision time

• Validity boundaries

Repeated decisions shall remain distinguishable.

Same Software Artifact ≠ Same Acceptance Decision.

Acceptance Decision Basis

QCAM shall preserve the evidentiary and methodological basis supporting

each acceptance decision.

The basis may include:

• Quality requirements

• Applicable criteria

• Conformity determination

• Intended-use validation

• Testing evidence

• Defect states

• Residual limitations

• Competent authority

A decision without sufficient supporting evidence shall remain

distinguishable from evidence-supported acceptance.

Recorded Acceptance Decision ≠ Justified Acceptance Decision.

Acceptance Decision Outcomes

QCAM shall support identifiable acceptance outcomes.

Possible outcomes include:

Accepted: The software quality state is accepted within defined conditions and scope.

Conditionally Accepted: Acceptance is granted subject to explicit limitations or prerequisites.

Rejected: The quality state is not accepted for the relevant purpose.

Deferred: The acceptance decision is postponed pending additional evidence or action.

Suspended: A previously recognized acceptance is temporarily not relied upon.

Withdrawn: A previously recognized acceptance has been revoked by

competent authority.

Superseded: A later decision has replaced the earlier acceptance

reference.

Unresolved: Available information does not support a sufficiently determined outcome.

Decision outcomes shall remain distinguishable from technical conformity

states.

Accepted Quality State ≠ Universally Conforming Software.

Acceptance Conditions

QCAM shall preserve explicit conditions attached to quality acceptance.

Conditions may concern:

• Supported software version

• Required configuration

• Intended-use restrictions

• Workload limitations

• Known defects

• Required monitoring

• Required corrective action

• Reassessment triggers

• Time-limited validity

Acceptance conditions shall remain identifiable and enforceable through competent downstream governance.

Acceptance with Conditions ≠ Unrestricted Acceptance.

Conditional Quality Acceptance

QCAM shall support quality acceptance under explicitly bounded conditions

where competent authority permits it.

Conditional acceptance may be appropriate where:

• A known defect has limited impact

• A noncritical quality criterion remains partially satisfied

• A specific configuration is verified

• An intended-use restriction is established

• Additional evidence is required before broader acceptance

Conditional acceptance shall not conceal unresolved critical quality

conditions.

Conditional Quality Acceptance ≠ Complete Quality Conformity.

Acceptance Exclusions

QCAM shall identify software functions, configurations, environments, or

use cases excluded from acceptance.

Exclusions may include:

• Unsupported functionality

• Untested integrations

• Unverified platforms

• Specific workload conditions

• Known problematic configurations

• Experimental features

Exclusions shall remain distinguishable from accepted scope.

Acceptance Exclusion ≠ Accepted Quality Condition.

Acceptance Restrictions

QCAM shall preserve restrictions governing reliance on an accepted quality

state.

Restrictions may concern:

• Permitted use

• Operational scale

• Supported users

• Workload

• Environment

• Duration

• Required safeguards

A restriction shall not be interpreted as evidence that the underlying software defect or limitation has been eliminated.

Operational Restriction ≠ Technical Defect Resolution.

Acceptance Validity

QCAM shall establish the conditions under which a quality acceptance

decision remains valid.

Validity may depend on:

• Artifact identity

• Configuration

• Intended use

• Applicable criteria

• Supporting evidence

• Defect conditions

• Operational assumptions

• Decision authority

Acceptance shall not be treated as permanently valid independent of

material changes.

Previously Accepted Quality ≠ Permanently Accepted Quality.

Temporal Acceptance Boundaries

QCAM shall preserve time-related limitations affecting quality acceptance.

These may include:

• Acceptance date

• Relevant evidence date

• Time-limited conditions

• Review requirements

• Reassessment deadlines

• Supersession events

The passage of time alone shall not necessarily invalidate acceptance.

However, material changes or expired acceptance conditions may require reassessment.

Historical Acceptance ≠ Current Acceptance Validity.

Acceptance Evidence Governance

QCAM shall preserve competent evidence supporting acceptance decisions.

Evidence may include:

• Conformity assessment

• Quality criteria references

• Testing and coverage records

• Defect governance states

• Intended-use validation

• Residual limitation records

• Authority references

• Decision records

Evidence shall remain connected to the actual acceptance subject.

Acceptance Documentation ≠ Acceptance Evidence Sufficiency.

Acceptance Evidence Competence

QCAM shall assess whether the evidence supporting acceptance is suitable for the decision being made.

Competence may depend on:

• Relevant software artifact

• Applicable criteria

• Suitable testing evidence

• Defect verification

• Intended-use correspondence

• Known limitations

• Valid authority

Evidence sufficient for a limited component acceptance may be insufficient for an integrated system acceptance.

Competent Component Evidence ≠ Competent System Acceptance Evidence.

Acceptance Evidence Integrity

QCAM shall preserve the integrity of material acceptance evidence.

Relevant considerations may include:

• Evidence source

• Artifact correspondence

• Record integrity

• Decision identity

• Temporal relevance

• Material contradictions

• Preservation of historical records

An intact acceptance record shall not independently establish that the decision was competent.

Intact Acceptance Record ≠ Valid Quality Acceptance.

Acceptance Conflict

An Acceptance Conflict exists where material acceptance evidence, authority references, conditions, or decisions are incompatible.

Examples may include:

• Conflicting conformity assessments

• Contradictory defect states

• Incompatible acceptance decisions

• Conflicting authority claims

• Contradictory artifact references

• Inconsistent intended-use conditions

QCAM shall preserve the conflict and its supporting evidence.

Conflicting Acceptance Evidence ≠ Permission to Select the Preferred

Decision.

Acceptance Evidence Gap

An Acceptance Evidence Gap exists where material information required for competent acceptance determination remains unavailable or insufficient.

Examples may include:

• Missing testing evidence

• Unverified artifact identity

• Incomplete coverage assessment

• Unresolved defect verification

• Missing intended-use validation

• Unclear acceptance authority

• Missing residual limitation assessment

An evidence gap shall not automatically establish nonconformity.

Acceptance Evidence Gap ≠ Proven Software Nonconformity.

Critical Acceptance Gap

A Critical Acceptance Gap exists where missing or unresolved evidence prevents competent acceptance determination for a materially consequential quality condition.

Examples may include:

- A critical financial calculation without sufficient quality

verification. - A safety-relevant function without competent intended-use validation. - An unresolved high-impact defect whose correction has not been verified. - An acceptance decision lacking identifiable authority. - A release candidate whose tested artifact identity is unresolved.

QCAM shall preserve the critical gap for competent governance.

A critical acceptance gap prevents an unsupported positive acceptance

determination for the affected scope; it does not independently establish that the software has failed or that release must be rejected under another standard.

Critical Acceptance Gap ≠ Automatic Software Failure Determination.

Acceptance Continuity

QCAM shall preserve acceptance relationships across software lifecycle

changes.

Continuity may involve:

• Historical acceptance decisions

• Artifact revisions

• Configuration changes

• Updated quality criteria

• New defect evidence

• Intended-use changes

• Authority changes

• Reassessment history

Historical acceptance records shall remain identifiable.

Acceptance Record Continuity ≠ Automatic Acceptance Continuation.

Acceptance Reassessment

QCAM shall support reassessment when material conditions affecting acceptance change.

Triggers may include:

• New software versions

• Configuration changes

• Dependency changes

• Revised quality requirements

• Updated quality criteria

• Newly discovered defects

• Failed defect verification

• New intended-use conditions

• Contradictory evidence

• Changes in acceptance authority

Reassessment shall preserve relevant historical determinations.

Acceptance Reassessment ≠ Historical Decision Erasure.

Acceptance Suspension

QCAM shall recognize temporary suspension of reliance on an acceptance

decision.

Suspension may arise where:

• Material evidence becomes unreliable

• Critical quality conditions become unresolved

• Acceptance conditions are violated

• Relevant software correspondence is lost

• Competent authority requires reassessment

Suspension shall remain distinguishable from withdrawal.

Acceptance Suspension ≠ Acceptance Withdrawal.

Acceptance Withdrawal

QCAM shall recognize withdrawal of previously granted quality acceptance by

competent authority.

Withdrawal may arise from:

• Material nonconformity

• Invalid supporting evidence

• Significant defect findings

• Invalid acceptance authority

• Loss of essential acceptance conditions

Withdrawal shall preserve the original decision and its historical basis.

Acceptance Withdrawal ≠ Erasure of Historical Acceptance.

Acceptance Supersession

QCAM shall preserve relationships between earlier and later acceptance

decisions.

A later acceptance may supersede an earlier decision where:

• A new artifact is assessed

• Criteria change

• Intended use changes

• Additional evidence becomes available

• Acceptance conditions are revised

Supersession shall remain explicit.

New Acceptance Decision ≠ Automatic Supersession of Every Earlier

Decision.

Quality Governance Handoff

QCAM shall preserve the information required for competent downstream

quality governance.

The handoff should identify:

• Accepted software object

• Artifact identity

• Configuration

• Applicable criteria

• Conformity determination

• Intended-use validation

• Testing evidence

• Defect conditions

• Residual limitations

• Acceptance authority

• Acceptance decision

• Conditions and exclusions

• Validity boundaries

• Reassessment triggers

Downstream release and deployment governance shall evaluate this

information under their own competent requirements.

Quality-Accepted ≠ Release-Authorized.

Conformity and Acceptance Governance States

QCAM shall support identifiable governance states.

Possible conformity states include:

Assessment-Pending: Conformity assessment has not been completed.

Assessment-Active: Conformity assessment is underway.

Conforming: Competent evidence supports applicable quality criteria.

Partially-Conforming: Only part of the applicable criteria is sufficiently satisfied.

Nonconforming: Competent evidence demonstrates non-satisfaction of

applicable criteria.

Inconclusive: Available evidence does not support a sufficient determination.

Evidence-Limited: Material evidence deficiencies remain.

Conflict-Detected: Material conformity evidence is incompatible.

Critical-Gap: A materially consequential quality condition lacks

sufficient evidence.

Possible acceptance states include:

Acceptance-Pending: Acceptance has not yet been determined.

Acceptance-Eligible: Conditions for competent acceptance consideration

are established.

Accepted: Competent authority has accepted the quality state within defined boundaries.

Conditionally-Accepted: Acceptance is subject to explicit conditions.

Rejected: Competent authority has not accepted the quality state.

Deferred: Acceptance consideration has been postponed.

Suspended: Reliance on an earlier acceptance is temporarily suspended.

Withdrawn: An earlier acceptance has been revoked.

Superseded: A later acceptance decision has replaced the earlier

reference.

Reassessment-Required: Material changes require renewed evaluation.

Unresolved: Available information does not support a sufficiently determined state.

Conformity states and acceptance states shall remain distinguishable.

Software may be technically conforming without an authorized acceptance decision.

A software quality state may also be conditionally accepted while known limitations remain explicitly governed.

Quality Conformity State ≠ Quality Acceptance State.

Governed Software Quality Conformity & Acceptance State

The principal output of QCAM is:

Governed Software Quality Conformity & Acceptance State

The state shall preserve sufficient information to identify:

Conformity Governance Object: The governed conformity assessment.

Acceptance Governance Object: The governed acceptance decision.

Assessment Identity: The identifiable quality assessment.

Assessment Purpose: The reason conformity is evaluated.

Assessment Scope: The applicable quality boundaries.

Software Object: The software being assessed.

Artifact Identity: The relevant software artifact.

Artifact Version: The assessed version.

Configuration Correspondence: The applicable configuration.

Quality Requirements: The governed requirements.

Quality Criteria: The applicable criteria.

Criteria Validity: The recognized applicability of the criteria.

Intended Use: The defined operational purpose.

Intended-Use Boundaries: The conditions covered.

Operational Context: The relevant operating conditions.

Quality Evidence Sufficiency: The competence of available evidence.

Testing Evidence: The relevant testing and coverage state.

Coverage Adequacy: The sufficiency of coverage for the claimed scope.

Defect State: The relevant governed defect conditions.

Known Defects: Material unresolved defects.

Residual Limitations: Known quality restrictions.

Conformity Determination: The recognized conformity outcome.

Partial Conformity: Material partially satisfied conditions.

Nonconformity: Demonstrated failures against criteria.

Conformity Uncertainty: Material unresolved evidence.

Intended-Use Validation: The relevant validation outcome.

Operational Fitness: The supported fitness determination.

Acceptance Eligibility: Whether acceptance consideration is sufficiently supported.

Acceptance Authority: The legitimate decision authority.

Authority Scope: The boundaries of acceptance authority.

Acceptance Decision Identity: The identifiable decision.

Acceptance Decision Basis: The supporting evidence and criteria.

Acceptance Outcome: The recognized decision.

Acceptance Conditions: Applicable prerequisites or limitations.

Conditional Acceptance: Relevant bounded acceptance conditions.

Acceptance Exclusions: Excluded software or operating conditions.

Acceptance Restrictions: Restrictions on reliance or use.

Acceptance Validity: The conditions under which acceptance remains

applicable.

Temporal Boundaries: Relevant dates and time limitations.

Acceptance Evidence: The evidence supporting the decision.

Evidence Competence: The suitability of that evidence.

Evidence Integrity: The condition of supporting evidence.

Acceptance Conflicts: Contradictory evidence or decisions.

Acceptance Evidence Gaps: Missing material evidence.

Critical Acceptance Gaps: Material unresolved conditions requiring

competent attention.

Acceptance Continuity: Preservation of lifecycle relationships.

Acceptance Reassessment: Material reassessment history.

Acceptance Suspension: Relevant suspended acceptance states.

Acceptance Withdrawal: Relevant withdrawn acceptance decisions.

Acceptance Supersession: Relationships between successive decisions.

Conformity Governance State: The recognized conformity condition.

Acceptance Governance State: The recognized acceptance condition.

Downstream References: Information required by parent SQS and competent

release or deployment governance.

The state shall not independently establish software security conformity,

release authorization, deployment authorization, legal compliance, or legal liability.

Minimum Implementation Framework

A conforming implementation of QCAM must provide the following six functional capabilities.

• Conformity Object, Software Identity, Quality Criteria, and Assessment

Scope Governance

Capability to establish identifiable conformity governance objects,

preserve software artifact and configuration correspondence, associate applicable quality requirements and criteria, and establish assessment purpose, scope, and intended-use boundaries.

The implementation must distinguish quality criteria applicability from demonstrated criterion satisfaction and prevent conformity claims from being generalized beyond the assessed software state.

• Testing, Coverage, Defect, Evidence Sufficiency, and Conformity

Determination Governance

Capability to receive governed testing and defect evidence, assess evidence

competence, evaluate coverage adequacy, recognize known defects and residual limitations, and establish supported conformity determinations.

The implementation must distinguish passing tests from quality conformity and partial conformity from complete conformity.

• Intended-Use Validation, Operational Fitness, and Acceptance Eligibility

Governance

Capability to establish intended-use conditions, assess operational

correspondence, identify relevant limitations, evaluate operational fitness, and determine whether the available quality state is eligible for acceptance consideration.

The implementation must distinguish technical criterion conformity from demonstrated fitness for intended use and acceptance eligibility from actual acceptance.

• Acceptance Authority, Decision Identity, Decision Basis, and Conditional

Acceptance Governance

Capability to preserve legitimate acceptance authority, establish authority

scope, identify acceptance decisions, preserve supporting evidence, record decision outcomes, and establish explicit conditions, exclusions, and restrictions.

The implementation must distinguish quality assessment competence from acceptance authority and conditional acceptance from unrestricted acceptance.

• Acceptance Evidence, Conflict, Gap, Validity, Continuity, and

Reassessment Governance

Capability to preserve competent and integrity-supported acceptance

evidence, recognize uncertainty, conflicts, evidence gaps and critical gaps, establish acceptance validity boundaries, preserve suspension, withdrawal, supersession, and reassessment history.

The implementation must prevent historical acceptance decisions from being treated as automatically valid after material software or operational changes.

• Governed Software Quality Conformity & Acceptance State and SQS Handoff

Governance

Capability to produce a Governed Software Quality Conformity & Acceptance

State with sufficient software identity, criteria, conformity evidence, intended-use validation, residual limitations, acceptance authority, decision conditions, validity boundaries, and continuity information for competent parent SQS and downstream lifecycle governance.

The implementation must not represent quality acceptance as proof of defect-free software or as independent authorization for software release or deployment. GIE™ Compatibility

QCAM is compatible with GIE™ as a downstream methodological analysis and execution layer.

GIE™ may support:

• Conformity governance object identification

• Acceptance governance object identification

• Software artifact correspondence analysis

• Configuration correspondence assessment

• Quality requirement mapping

• Quality criterion applicability analysis

- Quality criterion validity assessment
- Intended-use definition analysis
- Intended-use boundary identification
- Operational context mapping
- Quality evidence sufficiency assessment
- Testing evidence correspondence
- Coverage adequacy analysis
- Defect state correspondence
- Known defect assessment
- Residual quality limitation identification
- Quality conformity determination support
- Partial conformity identification
- Quality nonconformity recognition
- Conformity uncertainty detection
- Intended-use validation support
- Operational fitness analysis
- Acceptance eligibility assessment
- Acceptance authority reference analysis
- Acceptance authority scope assessment
- Acceptance decision identity mapping
- Acceptance decision basis evaluation
- Acceptance condition identification
- Conditional acceptance analysis
- Acceptance exclusion identification
- Acceptance restriction mapping
- Acceptance validity assessment
- Temporal acceptance boundary analysis
- Acceptance evidence competence assessment
- Acceptance evidence integrity assessment
- Acceptance conflict detection
- Acceptance evidence gap detection
- Critical acceptance gap detection
- Acceptance continuity mapping
- Acceptance reassessment support
- Acceptance suspension analysis
- Acceptance withdrawal analysis

• Acceptance supersession mapping

• Governed conformity and acceptance state representation

GIE™ may identify insufficient evidence, detect inconsistencies between testing results and acceptance claims, recognize missing acceptance authority, or propose reassessment.

Such outputs shall remain distinguishable from authorized acceptance

decisions.

GIE™ must not fabricate conformity evidence, invent acceptance authority, treat generated recommendations as binding acceptance decisions, erase residual limitations, or represent a technically favorable quality assessment as independent release or deployment authorization.

OOF® logic adapts to the governed reality. The engine adapts to OOF®

logic.

Governance Requirements

A conforming QCAM implementation shall satisfy the following requirements.

QCAM-R01 — Conformity Identity: Material conformity assessments shall

remain identifiable.

QCAM-R02 — Acceptance Identity: Material acceptance decisions shall

remain identifiable.

QCAM-R03 — Software Correspondence: Assessments shall identify the

software artifact and configuration concerned.

QCAM-R04 — Quality Requirement Correspondence: Conformity assessments

shall reference applicable governed quality requirements.

QCAM-R05 — Criteria Applicability: Applicable quality criteria shall

remain identifiable.

QCAM-R06 — Assessment Scope: Conformity and acceptance claims shall

preserve their assessment boundaries.

QCAM-R07 — Intended-Use Definition: Relevant intended-use conditions

shall be established.

QCAM-R08 — Evidence Sufficiency: Conformity determinations shall be

supported by competent evidence.

QCAM-R09 — Testing Correspondence: Relevant testing evidence shall remain

traceable.

QCAM-R10 — Coverage Adequacy: Material testing coverage limitations shall

be assessed.

QCAM-R11 — Defect Correspondence: Relevant defect governance states shall

remain identifiable.

QCAM-R12 — Residual Limitations: Material unresolved quality conditions

shall remain explicit.

QCAM-R13 — Conformity Determination: Quality conformity outcomes shall be

distinguishable.

QCAM-R14 — Partial Conformity: Partial conformity shall not be

represented as full conformity.

QCAM-R15 — Intended-Use Validation: Technical conformity shall not

automatically establish intended-use fitness.

QCAM-R16 — Acceptance Eligibility: Acceptance eligibility shall be

distinguishable from acceptance decisions.

QCAM-R17 — Acceptance Authority: Material acceptance decisions shall have

identifiable legitimate authority.

QCAM-R18 — Authority Scope: Acceptance authority shall remain bounded.

QCAM-R19 — Decision Basis: Acceptance decisions shall preserve their

evidentiary basis.

QCAM-R20 — Conditional Acceptance: Acceptance conditions and restrictions

shall remain explicit.

QCAM-R21 — Acceptance Exclusions: Excluded software or operational

conditions shall remain identifiable.

QCAM-R22 — Acceptance Validity: Material validity boundaries shall be

preserved.

QCAM-R23 — Evidence Integrity: Supporting acceptance evidence shall

preserve relevant integrity conditions.

QCAM-R24 — Acceptance Conflict: Material contradictory evidence or

decisions shall remain visible.

QCAM-R25 — Acceptance Evidence Gap: Missing material evidence shall not

be treated as demonstrated conformity.

QCAM-R26 — Critical Acceptance Gap: Material unresolved quality

conditions shall prevent unsupported positive acceptance determinations for the affected scope.

QCAM-R27 — Acceptance Continuity: Acceptance history shall remain

traceable across lifecycle changes.

QCAM-R28 — Acceptance Reassessment: Material changes shall trigger

competent reassessment where applicable.

QCAM-R29 — Acceptance Withdrawal: Suspension, withdrawal, and

supersession shall remain distinguishable.

QCAM-R30 — Governance Handoff: Conformity and acceptance states shall

preserve information required for competent downstream governance.

Governance Logic

The canonical QCAM governance logic is:

Receive Governed Software Quality Requirements State

↓

Receive Governed Quality Criteria Definition State

↓

Receive Governed Software Testing & Coverage State

↓

Receive Governed Software Defect State

↓

Identify Quality Conformity Governance Object

↓

Identify Software Object, Artifact, Version, and Configuration

↓

Establish Quality Assessment Purpose and Scope

↓

Identify Applicable Quality Requirements and Criteria

↓

Assess Quality Criteria Validity

↓

Define Intended Use and Operational Boundaries

↓

Assess Testing Evidence Correspondence

↓

Assess Testing Coverage Adequacy

↓

Assess Defect Governance States

↓

Identify Known Defects and Residual Limitations

↓

Assess Quality Evidence Sufficiency

↓

Determine Quality Conformity State

↓

Assess Intended-Use Validation

↓

Assess Operational Fitness

↓

Determine Acceptance Eligibility

↓

Identify Quality Acceptance Governance Object

↓

Establish Acceptance Authority and Authority Scope

↓

Preserve Acceptance Decision Basis

↓

Determine Acceptance Outcome

↓

Establish Acceptance Conditions, Exclusions, and Restrictions

↓

Preserve Acceptance Evidence and Integrity

↓

Identify Acceptance Conflicts and Evidence Gaps

↓

Establish Acceptance Validity Boundaries

↓

Determine Conformity and Acceptance Governance States

↓

Establish Governed Software Quality Conformity & Acceptance State

↓

Handoff to Parent SQS Governance

↓

Provide Quality References to Competent Release and Deployment Governance

↓

Preserve Acceptance Continuity

↓

Reassess When Material Quality or Operational Conditions Change

The principal output is:

Governed Software Quality Conformity & Acceptance State

Governance Outputs

QCAM may produce:

Quality Conformity Governance Object

Quality Acceptance Governance Object

Quality Assessment Identity Record

Quality Assessment Purpose Record

Quality Assessment Scope Record

Software Object Correspondence Record

Assessed Artifact Identity Record

Artifact Version Correspondence Record

Configuration Correspondence Record

Quality Requirement Conformity Record

Quality Criteria Applicability Record

Quality Criteria Validity Assessment

Intended-Use Definition Record

Intended-Use Boundary Record

Operational Context Record

Quality Evidence Sufficiency Assessment

Testing Evidence Correspondence Record

Coverage Adequacy Assessment

Defect State Correspondence Record

Known Defect Assessment

Residual Quality Limitation Record

Quality Conformity Determination

Partial Quality Conformity Record

Quality Nonconformity Record

Conformity Uncertainty State

Intended-Use Validation Record

Operational Fitness Assessment

Quality Acceptance Eligibility Record

Acceptance Authority Record

Acceptance Authority Scope Record

Acceptance Decision Identity Record

Acceptance Decision Basis Record

Acceptance Decision Outcome

Acceptance Conditions Record

Conditional Acceptance Record

Acceptance Exclusion Record

Acceptance Restriction Record

Acceptance Validity Record

Temporal Acceptance Boundary Record

Acceptance Evidence Register

Acceptance Evidence Competence Assessment

Acceptance Evidence Integrity Assessment

Acceptance Conflict

Acceptance Evidence Gap

Critical Acceptance Gap

Acceptance Continuity Record

Acceptance Reassessment Record

Acceptance Suspension Record

Acceptance Withdrawal Record

Acceptance Supersession Record

Conformity Governance State

Acceptance Governance State

SQS Quality Conformity & Acceptance Handoff Record

The principal output is:

Governed Software Quality Conformity & Acceptance State

Critical Distinctions

QCAM preserves the following distinctions:

Quality Acceptance ≠ Proof of Defect-Free Software

Quality Conformity Determination ≠ Quality Acceptance Decision

Documented Acceptance ≠ Evidence-Supported Acceptance

Assessment Purpose ≠ Assessment Outcome

Quality Conformity Within Scope ≠ Universal Software Conformity

Component Quality Acceptance ≠ System Quality Acceptance

Declared Accepted Version ≠ Verified Accepted Artifact

Quality Acceptance of Version A ≠ Quality Acceptance of Version B

Accepted Configuration ≠ Every Supported Configuration

Requirement Marked Satisfied ≠ Demonstrated Requirement Satisfaction

Quality Criterion Applicability ≠ Quality Criterion Satisfaction

Established Criterion ≠ Currently Applicable Criterion

Declared Intended Use ≠ Demonstrated Intended-Use Fitness

Accepted Intended Use ≠ Unrestricted Software Use

Technical Test Success ≠ Demonstrated Operational Fitness

Valid Quality Evidence ≠ Sufficient Quality Evidence

Passing Test Report ≠ Quality Conformity Determination

High Test Coverage ≠ Sufficient Quality Evidence

Closed Defect Record ≠ Verified Quality Conformity

Known Defect Existence ≠ Automatic Quality Rejection

Residual Limitation Disclosure ≠ Residual Limitation Acceptance

Demonstrated Quality Conformity ≠ Authorized Quality Acceptance

Partial Quality Conformity ≠ Full Quality Conformity

Quality Nonconformity ≠ Automatically Rejected Acceptance Decision

Unresolved Quality Evidence ≠ Demonstrated Conformity

Technical Quality Conformity ≠ Intended-Use Validation

Fitness for One Operational Context ≠ Fitness for Every Operational

Context

Acceptance Eligibility ≠ Acceptance Authorization

Quality Assessment Competence ≠ Quality Acceptance Authority

Authority to Accept One Scope ≠ Authority to Accept Every Scope

Human Approval ≠ Competent Quality Acceptance

Automated Acceptance Recommendation ≠ Authorized Acceptance Decision

Same Software Artifact ≠ Same Acceptance Decision

Recorded Acceptance Decision ≠ Justified Acceptance Decision

Accepted Quality State ≠ Universally Conforming Software

Acceptance with Conditions ≠ Unrestricted Acceptance

Conditional Quality Acceptance ≠ Complete Quality Conformity

Acceptance Exclusion ≠ Accepted Quality Condition

Operational Restriction ≠ Technical Defect Resolution

Previously Accepted Quality ≠ Permanently Accepted Quality

Historical Acceptance ≠ Current Acceptance Validity

Acceptance Documentation ≠ Acceptance Evidence Sufficiency

Competent Component Evidence ≠ Competent System Acceptance Evidence

Intact Acceptance Record ≠ Valid Quality Acceptance

Conflicting Acceptance Evidence ≠ Permission to Select the Preferred

Decision

Acceptance Evidence Gap ≠ Proven Software Nonconformity

Critical Acceptance Gap ≠ Automatic Software Failure Determination

Acceptance Record Continuity ≠ Automatic Acceptance Continuation

Acceptance Reassessment ≠ Historical Decision Erasure

Acceptance Suspension ≠ Acceptance Withdrawal

Acceptance Withdrawal ≠ Erasure of Historical Acceptance

New Acceptance Decision ≠ Automatic Supersession of Every Earlier Decision

Quality-Accepted ≠ Release-Authorized

Quality Conformity State ≠ Quality Acceptance State

Cross-Architecture Boundary

QCAM governs software-quality-specific conformity assessment, intended-use

validation, acceptance eligibility, acceptance authority, acceptance decisions, residual limitations, validity, and continuity within SQS and SLGA™.

It does not independently establish quality requirements, define quality criteria, execute testing activities, resolve defects, certify security, authorize release, authorize deployment, or determine legal compliance.

Within SQS:

QRM — Quality Requirements Module governs software quality requirement

establishment.

QCDM — Quality Criteria Definition Module governs quality criterion

definition, applicability, assessment methods, and evidence requirements.

TCMM — Testing & Coverage Management Module governs testing activities,

execution evidence, observed results, and coverage.

DFM — Defect Management Module governs defect identification,

classification, impact, correction references, resolution verification, and continuity.

QCAM — Quality Conformity & Acceptance Module governs conformity

assessment, intended-use validation, acceptance authority, acceptance decisions, residual limitations, and acceptance continuity.

Within SLGA™:

Software Requirements Standard (SRS) governs the broader software

requirements lifecycle.

Software Creation Standard (SCS) governs software creation.

Software Dependency Standard (SDS) governs software dependencies.

Software Configuration Standard (SCFS) governs software configuration.

Software Quality Standard (SQS) governs software quality.

Software Security Standard (SSS) governs software security.

Software Release Standard (SRLS) governs software release.

Software Deployment Standard (SDPS) governs software deployment.

Software Change Management Standard (SCMS) governs broader lifecycle

changes.

Software Retirement Standard (SRTS) governs software retirement.

Software Lifecycle Control Standard (SLCS) governs cross-cutting

lifecycle control.

Relevant interoperating OOF® architectures include:

GOA™ — Governance Architecture: General authority and delegation

governance.

AGA™ — Accountability Governance Architecture: Relevant responsibility

and accountability relationships.

ORA™ — Operational Reality Architecture: Competent assessment of actual

software behavior and operating conditions.

ASGA™ — Autonomous Systems Governance Architecture: Relevant autonomous

assessment and decision activities.

AGI™ — Agent Governance Infrastructure: Relevant agent permissions,

tool-path authority, and constituent action governance.

QCAM preserves software-quality-specific conformity and acceptance

governance without replacing these competent domains.

Canonical Boundary

QCAM begins when a defined software quality state requires conformity assessment, intended-use validation, acceptance eligibility determination, acceptance decision, or reassessment.

QCAM governs:

• Quality conformity governance objects

• Quality acceptance governance objects

• Assessment identity

• Assessment purpose

• Assessment scope

• Software object correspondence

• Artifact identity

• Artifact versions

• Configuration correspondence

• Quality requirement correspondence

- Quality criteria applicability
- Quality criteria validity
- Intended-use definition
- Intended-use boundaries
- Operational context
- Quality evidence sufficiency
- Testing evidence correspondence
- Coverage adequacy
- Defect state correspondence
- Known defect assessment
- Residual quality limitations
- Quality conformity determinations
- Partial conformity
- Nonconformity
- Conformity uncertainty
- Intended-use validation
- Operational fitness
- Acceptance eligibility
- Acceptance authority
- Acceptance authority scope
- Acceptance decision identity
- Acceptance decision basis
- Acceptance outcomes
- Acceptance conditions
- Conditional acceptance
- Acceptance exclusions
- Acceptance restrictions
- Acceptance validity
- Temporal acceptance boundaries
- Acceptance evidence
- Acceptance evidence competence
- Acceptance evidence integrity
- Acceptance conflicts
- Acceptance evidence gaps
- Critical acceptance gaps
- Acceptance continuity

• Acceptance reassessment

• Acceptance suspension

• Acceptance withdrawal

• Acceptance supersession

• Quality governance handoff

• Conformity and acceptance governance states

QCAM does not independently determine:

• Complete software requirements validity

• Software creation authorization

• Dependency suitability

• Configuration baseline establishment

• Quality requirement definition

• Quality criterion establishment

• Actual test execution

• Defect root-cause attribution

• Defect correction authorization

• Defect resolution implementation

• Software security conformity

• Release authorization

• Deployment authorization

• Legal compliance

• Legal liability

QCAM ends when material conformity and acceptance conditions have been represented as a:

Governed Software Quality Conformity & Acceptance State

with sufficient software identity, quality criteria, testing and defect

evidence, intended-use validation, conformity determination, acceptance authority, acceptance conditions, residual limitations, validity boundaries, and continuity information for competent parent SQS and downstream lifecycle governance.

Use Case 1 — Financial Software Passes Quality Tests but Remains

Unvalidated for Its Intended Operational Workload

An organization develops software for processing financial transactions.

The software passes:

• Functional unit tests

• Integration tests

• Selected regression tests

• Defined performance tests

The testing team reports that all established quality criteria have been satisfied.

However, the intended operational environment requires substantially higher

transaction concurrency than the assessed environment.

The available performance tests do not represent the actual peak workload.

The software also contains a known intermittent processing defect whose

correction has not been verified under representative conditions.

Management proposes accepting the software because the existing quality

dashboard reports favorable results.

QCAM evaluates:

Quality Requirements → Applicable Criteria → Testing Evidence → Defect

States → Intended Use → Operational Fitness → Conformity → Acceptance Eligibility

The assessment identifies:

• Competent evidence supporting selected technical criteria

• Insufficient evidence for the intended peak workload

• An unresolved intermittent defect

• A material intended-use validation gap

• Insufficient support for unrestricted quality acceptance

The resulting governance state may recognize:

Partially-Conforming

for the assessed quality scope.

It may also recognize:

Evidence-Limited

for the intended operational workload.

A competent acceptance authority may defer acceptance or consider a narrower acceptance scope only where the remaining conditions are sufficiently established and permitted.

The critical distinctions are:

Technical Quality Conformity ≠ Intended-Use Validation.

High Test Coverage ≠ Sufficient Quality Evidence.

Acceptance Eligibility ≠ Acceptance Authorization.

Quality Acceptance ≠ Proof of Defect-Free Software.

The result: QCAM prevents successful testing under limited conditions

from being interpreted as unrestricted software quality acceptance for an operational environment that has not been sufficiently validated.

Use Case 2 — An Autonomous Development System Approves Its Own Software

Quality Without Legitimate Acceptance Authority

An organization uses an autonomous software development system to create

and test internal business software.

The system:

• Generates application code

• Executes automated tests

• Analyzes test coverage

• Reviews known defects

• Produces a quality assessment

• Marks the software as accepted

The automated report states:

"Quality verified. Approved for production."

However:

• The system was authorized to develop and test software

• It was not authorized to make final quality acceptance decisions

• Its testing evidence contains unresolved coverage exclusions

• A known defect remains conditionally unresolved

• No competent authority has accepted the residual limitations

• The report does not establish a defined acceptance scope

QCAM evaluates:

Conformity Evidence → Intended-Use Validation → Residual Limitations →

Acceptance Eligibility → Acceptance Authority → Decision Basis → Acceptance Outcome

The assessment identifies:

• A completed automated quality assessment

• Material unresolved quality limitations

• Insufficient acceptance authority

• No valid final acceptance decision

• An unsupported claim of production approval

The relevant governance state may be:

Acceptance-Pending

with:

Evidence-Limited

for the unresolved quality conditions.

The critical distinctions are:

Automated Acceptance Recommendation ≠ Authorized Acceptance Decision.

Quality Assessment Competence ≠ Quality Acceptance Authority.

Recorded Acceptance Decision ≠ Justified Acceptance Decision.

Quality-Accepted ≠ Release-Authorized.

The result: QCAM prevents an autonomous development system from

converting its own technical assessment into an acceptance decision beyond its legitimate authority, while preserving the quality evidence for competent human or organizational review.

Canonical Closing Statement

Software quality is not accepted merely because testing is complete, defects are closed, quality metrics appear favorable, or an automated system declares that the software is ready.

A software artifact may satisfy defined technical criteria while remaining insufficiently validated for its intended use.

A quality assessment may be technically competent while the actor issuing acceptance lacks legitimate authority.

A software system may be accepted for one configuration while remaining unverified for another.

A known defect may be compatible with explicitly bounded acceptance while an undisclosed residual limitation may invalidate the basis of that decision.

And a previously accepted software quality state may require reassessment when the artifact, configuration, intended use, or supporting evidence materially changes.

Quality Conformity Determination ≠ Quality Acceptance Decision.

Technical Quality Conformity ≠ Intended-Use Validation.

Acceptance Eligibility ≠ Acceptance Authorization.

Conditional Quality Acceptance ≠ Complete Quality Conformity.

Quality-Accepted ≠ Release-Authorized.

Quality Acceptance ≠ Proof of Defect-Free Software.

These distinctions become increasingly important as software creation,

testing, assessment, and decision support become more automated.

An autonomous system may generate convincing quality evidence.

It may identify apparent conformity.

It may recommend acceptance.

But the technical ability to produce an assessment does not independently establish legitimate authority to accept software quality or determine the conditions under which others may rely on that acceptance.

QCAM therefore governs the relationship between demonstrated quality

conditions, intended-use fitness, competent evidence, residual limitations, legitimate acceptance authority, and the continuing validity of acceptance decisions.

It does not require software to be free of every possible defect before acceptance can be considered.

It requires known limitations to remain explicit and acceptance decisions

to be bounded by competent evidence and legitimate authority.

It does not equate acceptance with release or deployment.

It establishes the quality-specific state upon which those separate

governance decisions may subsequently rely.

QCAM transforms software quality acceptance from a declaration, checklist

completion, management approval, or automated readiness status into a governed determination — ensuring that software organizations can establish which quality conditions have actually been demonstrated, which intended uses are sufficiently supported, what limitations remain, who is authorized to accept them, and under which conditions that acceptance remains valid.

Software quality is not governed because it has been declared acceptable. It is governed when conformity, intended-use validation, evidence sufficiency, residual limitations, acceptance authority, decision boundaries, and lifecycle continuity remain sufficiently established to support competent and accountable quality acceptance.